School of Engineering & Technology
MGMU, Chh. Sambhajinagar

Department of Computer Science & Engineering

Practical Workbook

Academic Year 2026-27

Class: Fifth Year Integrated Data Science (FI(I)CSE(DS))
Course Code: CSI24PCP501
Course Title: Data Story Telling & Visualization Lab
Semester-IX

Prepared by:
Ms. Sarika Rathi
Course Coordinator- Data Story Telling & Visualization Lab

Lab Outcomes: Students will be able to:

LO1: Practically identify and analyze the key elements and characters involved in data storytelling within real-time datasets.

LO2: Apply techniques to eliminate visual and narrative clutter, and construct compelling data-driven stories for given case scenarios.

LO3: Transform raw datasets into meaningful visual representations using appropriate data preprocessing and visualization tools.

LO4: Design and develop data visualizations using multiple distribution types (e.g., normal, binomial, Poisson, uniform, etc.) to interpret and communicate insights effectively.

LO-PO Mapping

(Mapping scale: 1:Low, 2:Medium, 3: High)

LO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO1												
LO2												
LO3												
LO4												

LO-PSO Mapping

(Mapping scale: 1:Low, 2:Medium, 3: High)

LO	PSO1	PSO2	PSO3
LO1			
LO2			
LO3			
LO4			
LO5			

*PO: Program Outcomes

*PSO: Program Specific Outcomes

(* Refer Appendix for details)

Practical Record

Name of student:_____

Roll No: _____ PNR No:_____

Batch: _____

Sr. No.	Practical details	Date of performance	Date of evaluation	Marks/ Grade	Sign

Submission Details:

Date: Sign of Faculty Sign of HOD

Practical No.1

Identify a project you are working on where you need to communicate in a data-driven way. Reflect upon and write the answers of following questions.

a) What is a Stake?

i) What are the benefits if your audience acts in the way that you want them to?

ii) What are the risks if they do not?

b) Form a Big Idea

i) Articulate your point of view.

ii) Convey what's at stake.

I. Practical Significance

By learning data storytelling, students can improve both their analytical and communication skills. It helps in presenting data in a simple and clear way so that people can understand it easily and take proper decisions. In real situations, it supports better planning, improves productivity, and helps different teams like management, HR, and team leaders to work more efficiently. This practical mainly helps students understand how to clearly explain what is important, what risks are involved, and what benefits can be achieved. It also helps in creating a strong and clear Big Idea, which makes communication more focused and effective. Overall, this exercise improves thinking, decision-making, and the ability to present ideas clearly using data.

II. Course Level Learning Outcome

CO1-Identify the characters data storytelling.

III. Laboratory Learning Outcome

1. Identify and analyze the target audience
2.Interpret audience needs
3.Define stake including risks and benefits
4.Construct a clear and actionable Big Idea

IV. Relevant Theoretical Background

Data storytelling is a structured approach to communicating insights derived from data using a combination of visuals, narrative, and context. Before creating any visualization, it is essential to understand the purpose and the impact of the message. One of the key concepts in this process is

identifying the stake, which refers to what is at risk or what can be gained depending on the decisions made by the audience. Clearly defining the stake helps in making the story relevant and persuasive.

Another important concept is understanding the audience. Different audiences require different levels of detail, tone, and presentation style. For example, technical audiences may prefer detailed data analysis, whereas business stakeholders may focus more on key insights and outcomes. Therefore, tailoring the message according to the audience is crucial for effective communication.

The Big Idea is the central message of the data story. It is a single, clear, and complete sentence that conveys the main point, highlights what is at stake, and suggests an action. A strong Big Idea should be concise, specific, and impactful. It acts as a guiding statement that ensures all visualizations and narratives remain focused and aligned with the objective.

In practical applications, these elements are widely used in business analytics, reporting dashboards, and decision-making processes. Without a clear Big Idea and understanding of stakeholders, data presentations often become confusing and fail to drive action. Hence, this practical emphasizes the importance of combining analytical thinking with structured communication.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Optional tools: Python/ Microsoft Power BI / Tableau

VI. Procedure with Example Scenario for Students: Analyzing Exam Results to Improve Academic Performance

Imagine a student working on a project to analyze exam results in a hypothetical high school. They gather data on various factors such as study habits, attendance records, extracurricular involvement, and previous academic performance. Using data Storytelling, the student can: Create a Narrative: Develop a story that shows a correlation between consistent study habits and higher exam scores. The narrative can highlight how students who maintain regular study schedules tend to perform better academically.

Use Visual Aids: Utilize visual aids like scatter plots, bar graphs, and pie charts to illustrate the

impact of different factors on exam performance. For example, a scatter plot can show the relationship between hours spent studying and exam scores, while a pie chart can depict the distribution of scores across different attendance levels.

Present Recommendations: Based on the data analysis, the student can recommend strategies such as implementing study skill workshops, offering after-school tutoring programs, or encouraging regular attendance through incentive programs.

Highlight ROI: Demonstrate the potential return on investment (ROI) of these recommendations. For instance, the student can calculate the expected improvement in overall school performance and the long-term benefits of better academic outcomes, such as higher college acceptance rates and scholarships.

VII. Multiple-choice questions (MCQs)

1. What is the primary purpose of identifying stakeholders in data storytelling?

A) To select visualization tools

B) To understand who will take action based on insights

C) To clean data

D) To store data

2. A strong Big Idea should include:

A) Multiple charts

B) Only raw data

C) Clear message and action

D) Code implementation

3. What does “risk” represent in storytelling?

A) Positive outcome

B) Possible negative consequence if no action is taken

C) Data type

D) Visualization style

4. Why is audience analysis important?

A) It reduces dataset size

B) It helps tailor communication effectively

C) It improves coding speed

D) It removes errors

5. Which of the following best describes a Big Idea?

A) A dataset

B) A detailed report

C) A single clear statement conveying insight and action

D) A chart type

VIII. Exercises

Follow the procedure given below:

Project Choice: Pick a topic for your project where data can help to tell a story.

1. **Understand Audience:** Know who you want to share the story with.
2. **Identify Stakeholders:** List the key individuals or groups involved or affected by the data story.
3. **Audience Needs:** Find out what information your audience wants.
4. **Identify Risks:** Determine what negative outcomes may occur if no action is taken.
5. **Identify Benefits:** Explain the positive outcomes if the audience takes the desired action.
6. **Form Big Idea:** Write a single clear sentence that conveys the main message and action.
7. **Story Design:** Create a clear story that fits what your audience needs.

Select any one choice and complete the activity.

Low Student Attendance in College

Traffic Congestion in Urban Areas

Hospital Patient Waiting Time Increase

Electricity Consumption Increase in a City

Delay in Delivery Services (E-commerce/Logistics)

IX. Practical Related Questions

1. Who is your audience?
2. What does your audience care about?
3. What is data story telling?
4. Write Risk and Benefits?
5. Write a code for Basic Data Analysis for Story Identification which includes Dataset, load the Dataset, clean the Dataset, do basic analysis of Dataset, find the insights, do simple visualization?

Practical No.2

Determine audience, communication mechanism, desired tone and Select effective Visuals for any incident. (Below is an example for reference only).

i) Who is your Audience?

ii) List the primary groups or individuals to whom you'll be communicating.

iii) If you had to narrow that to a single person, who would that be?

iv) What does your audience care about?

v) What action does your audience need to take?

Example: Buses Bunching - Make a data story on bus bunching, (Bus Bunching means when a bus gets delayed and later causes multiple buses to arrive at a single stop at the same time.

I. Practical Significance

This practical helps students understand how to communicate data effectively by identifying the audience and selecting the right way to present information. In real-life situations, even correct data may not be useful if it is not presented according to the audience's needs. Different people like managers, technical experts, or general users understand data in different ways, so the communication method must be chosen carefully.

Through this practical, students learn how to identify their audience, understand what they care about, and decide the best way to present data using reports, dashboards, or presentations. It also focuses on choosing the right tone and selecting suitable visuals like tables, bar charts, or graphs to make the information clear and easy to understand. Overall, this activity helps students connect data analysis with communication and improves their ability to present insights in a simple and effective way.

II. Course Level Learning Outcome

CO1-Identify the characters in data storytelling.

III. Laboratory Learning Outcome

1. Construct data story on a given incident.
2. Identify and analyze the target audience.
3. Select appropriate communication mechanism (report/dashboard/presentation).
4. Choose suitable tone for communication.
5. Select effective visualization.

IV. Relevant Theoretical Background

Data storytelling is not only about presenting data but also about communicating insights effectively to a specific audience**.** One of the most important steps in this process is identifying the audience. The audience can vary from technical experts to business stakeholders, and each group has different expectations, knowledge levels, and interests. Understanding what the audience cares about helps in selecting the right content and level of detail.

Another key concept is the communication mechanism, which refers to how the information is delivered. This can include dashboards, reports, presentations, or interactive visualizations. The choice of mechanism depends on the purpose of communication and the audience's preferences. For example, executives may prefer dashboards for quick insights, while analysts may require detailed reports.

The tone of communication is equally important. It can be formal, persuasive, or explanatory depending on the context. A persuasive tone is useful when the goal is to drive action, while an informative tone is suitable for reporting data insights. Selecting the right visualization is critical for effective storytelling. Simple visuals such as tables, bar charts, and line graphs should be chosen based on the type of data and the message to be conveyed. Poor visualization choices can confuse the audience, while effective visuals improve understanding and engagement. Tools such as Tableau and Microsoft Power BI are widely used to create interactive and professional visualizations in real-world applications.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Optional tools: Python/ Microsoft Power BI / Tableau

VI. Multiple-choice questions (MCQs)

1. Why is audience identification important in data storytelling?

A) To increase dataset size

B) To tailor communication effectively

C) To reduce data

D) To improve coding

2.Which communication mechanism is best for quick decision-making?

A) Long report

B) Dashboard

C) Raw data

D) Text file

3. What does tone represent in storytelling?

A) Color of chart

B) Style of communication

C) Dataset size

D) Tool used

4. Which visualization is best for comparing categories?

A) Pie chart

B) Bar chart

C) Text

D) Table

5. What is the main goal of selecting effective visuals?

A) Decoration

B) Data storage

C) Clear communication

D) Coding

VII. Exercises

Project Choice: Pick a real-life scenario where data can help explain a problem.

1. **Identify Audience:** Identify primary and secondary audience for your data story.
2. **Audience Analysis:** Understand what your audience cares about and what action they should take.
3. **Select Communication Mechanism:** Decide whether to use report, dashboard, or presentation based on audience.
4. **Decide Tone:** Choose appropriate tone (informative, persuasive, formal) for

communication.

5. **Select Visualization:** Choose suitable charts (bar, line, pie, table) based on data and message.
6. **Create Data Story:** Combine all elements to create a clear and meaningful data story.

Additional Real-World Scenarios (Choose Any One) and complete activity:

1. Decline in product sales
2. Increase in hospital waiting time
3. Traffic congestion
4. Student performance analysis
5. Online app usage trends
6. Employee performance review
7. Electricity consumption analysis
8. Bus bunching problem
9. Customer satisfaction survey
10. Website traffic analysis

VIII. Practical Related Questions

1.Explain how audience type influences visualization selection.

2. Compare dashboard and report as communication mechanisms.

3. Why is tone important in data storytelling?

4. How does incorrect visualization affect decision-making?

5. What factors should be considered while selecting visuals?

6. Write a code to check how visualization changes based on audience, which includes Dataset, load the Dataset, clean the Dataset, do basic analysis of Dataset, do simple visualization minimum 4 different methods and based on above decide which one is better?

Practical No.3

Make a clutter free Data Story on any incident.

a) Identify and eliminate clutter.

b) Select suitable pre-attentive attributes.

c) Explore affordances, accessibility, and aesthetics as per designer's view.

I. Practical Significance

Creating a clutter-free data story holds significant practical importance across various domains such as business, academia, journalism, and public policy. It enhances communication by presenting information clearly and efficiently, making key insights more accessible and understandable. This clarity supports informed decision-making, increases viewer engagement, and encourages exploration of the data without overwhelming the audience. Additionally, a clutter- free approach improves accessibility for diverse audiences and enhances the professional presentation of information, ultimately boosting credibility and trust in the data presented.

II. Course Level Learning Outcome

CO2-Eliminate clutter to grab audience attention.

CO3-Construct Story telling for the given incident.

III. Laboratory Learning Outcome

1. Construct a data story on a given incident.
2. Identify and eliminate clutter in visualizations
3. Choose suitable pre-attentive attributes.
4. Improve clarity and design from a user perspective.

IV. Relevant Theoretical Background

Clutter in data visualization refers to unnecessary or distracting elements that make it difficult to understand the main message. These elements may include excessive labels, too many colors, gridlines, 3D effects, or unnecessary text. Clutter increases cognitive load, which means the audience has to spend more effort to understand the information.

Decluttering is the process of removing these unnecessary elements to improve clarity and focus. A good visualization should highlight only the important information and avoid distractions. Pre-attentive attributes such as color, size, and position help in quickly drawing attention to key insights without requiring detailed analysis.

A well-designed visualization also considers accessibility and aesthetics. It should be easy to read, visually balanced, and understandable for all users. By reducing clutter and improving design, data storytelling becomes more effective, clear, and impactful.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Optional tools: Python/ Microsoft Power BI / Tableau

VI. Multiple-choice questions (MCQs)

1.What is clutter in data visualization?

A) Useful data

B) Unnecessary elements in a chart

C) Dataset

D) Algorithm

2.Cognitive load refers to:

A) Data storage

B) Mental effort required to understand data

C) Chart size

D) Coding complexity

3.Which of the following is an example of clutter?

A) Clear labels

B) Highlighted data

C) Excessive gridlines

D) Proper chart title

4.What is the main goal of decluttering?

A) Increase complexity

B) Improve clarity

C) Add more colors

D) Increase data size

5.Which technique helps in decluttering?

A) Adding 3D effects

B) Using multiple fonts

C) Removing unnecessary elements

D) Increasing chart size

VII. Exercises

Follow the procedure given below:

Project Choice: Select a real-life scenario (e.g., sales data, attendance, traffic).

1. **Create Cluttered Visualization:** Create or select a chart with unnecessary elements (colors, labels, gridlines).
2. **Identify Clutter:** List all unnecessary elements present in the visualization.
3. **Apply Decluttering:** Remove unwanted elements and simplify the design.
4. **Apply Pre-attentive Attributes:** Use color, size, or position to highlight important data.
5. **Improve Design:** Ensure clarity, readability, and proper alignment.
6. **Compare Results:** Compare cluttered and decluttered visualization and explain improvements.

Additional Real-World Scenarios (Choose Any One)

1. Sales dashboard with too many colors
2. Student result chart with excessive labels
3. Traffic data with unnecessary gridlines
4. Hospital data with confusing layout
5. E-commerce dashboard overloaded with visuals
6. Employee performance chart with clutter
7. Weather data chart with background images
8. Financial report with too much text
9. Mobile app analytics dashboard
10. Electricity usage graph.

VIII. Practical Related Questions

1. Explain how clutter affects decision-making in data visualization.
2. What is cognitive load and how does it impact understanding?

3. Describe the steps involved in decluttering a visualization.
4. Why are 3D charts often misleading?
5. How do pre-attentive attributes improve data storytelling?
6. Write a code to show difference between cluttered vs clean visualization includes Dataset, load the Dataset, clean the Dataset, do basic analysis of Dataset, Create CLUTTERED Chart, Create CLEAN Chart.?

Practical No.4

Create a data story for Vehicle (Bicycle/Bike/Car/Bus etc.) Rental System.

Example: Create a data story with respect to the following observations:

i. What are the most popular pick-up locations across the city for Citi Bike rental?
ii. How does the average trip duration vary across different age groups, and over time?
iii. Which age group rents the most bikes?
iv. How does bike rental vary across the two user groups (one-time users vs. long-term subscribers) on different days of the week?
v. Do factors like weather and user age impact the average bike trip duration?

I. Practical Significance

This practical focuses on applying data storytelling concepts to a real-world scenario of a vehicle rental system. In today's data-driven world, rental services such as bicycles, bikes, cars, and buses generate large amounts of data related to usage patterns, customer preferences, peak hours, and operational efficiency. However, raw data alone cannot provide meaningful insights unless it is transformed into a clear and structured story.

From this practical analysis, students will gain hands-on experience in data-driven decision-making within the context of a bike-sharing system. They will learn to identify key usage patterns and trends, understand user demographics and preferences, and evaluate the impact of external factors like weather on service usage. By engaging with real-world data, students will develop skills in data analysis, visualization, and interpretation, enabling them to draw meaningful conclusions that can inform operational and strategic decisions in transportation services and beyond.

II. Course Level Learning Outcome

CO1- Identify the characters in data Storytelling.

CO2- Eliminate clutter and grab audience attention.

CO3- Construct Story telling for the given incident.

III. Laboratory Learning Outcome

1. Construct a data story based on rental system data.
2. Analyze usage patterns and user behavior.
3. Identify key insights from the dataset.
4. Select appropriate visualizations for storytelling.

5. Present findings in a structured and meaningful way.

IV. Relevant Theoretical Background

Data storytelling is the process of converting raw data into meaningful insights using visuals and narrative. In a vehicle rental system, data helps in understanding patterns such as popular locations, peak usage times, user behavior, and trip duration.

By analyzing this data, we can identify trends like which locations have high demand, which age group uses the service more, and how factors like time and weather affect usage. These insights help in improving service quality, optimizing resource allocation, and making better decisions.

A good data story should include a clear Big Idea, proper visualizations, logical flow, and actionable insights. Instead of just showing data, the goal is to explain what is happening, why it is happening, and what should be done next.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Python/ Microsoft Power BI / Tableau

VI. Multiple-choice questions (MCQs)

1.What is the main goal of a data story in a rental system?

A) Store data

B) Communicate insights

C) Increase data size

D) Coding

2.Which visualization is best for showing rental trends over time?

A) Pie chart

B) Line chart

C) Table

D) Text

3. Who can be the audience for rental system data?

A) Only developers

B) Only customers

C) Managers, operators, customers

D) None

4. What is a Big Idea in storytelling?

A) Dataset

B) Chart

C) Main message

D) Code

5. Which chart is suitable for comparing vehicle types?

A) Line chart

B) Bar chart

C) Text

D) Image

VII. Exercises

Follow the procedure given below:

Project Choice: Select a dataset related to vehicle rental system.

1. **Understand Dataset**: Identify variables such as location, age, trip duration, user type.
2. **Identify Key Questions:** Define what you want to analyze (e.g., popular locations, age trends).
3. **Analyze Data:** Observe patterns and trends in the dataset.
4. **Select Visualization**: Choose appropriate charts (bar, line, pie).
5. **Create Data Story:** Combine insights into a clear story.
6. **Write Conclusion:** Provide recommendations based on analysis.

- **Additional Real-World Scenarios (Choose Any One)**

1. Bicycle rental demand in smart city
2. Car rental service performance
3. Bus ticket booking trends

4. Electric vehicle rental usage
5. Ride-sharing service demand
6. Weekend vs weekday rentals
7. Seasonal rental trends
8. Customer preference analysis
9. Fleet utilization efficiency
10. Rental pricing impact analysis

VIII. Practical Related Questions

1. Explain how data storytelling improves decision-making in rental systems.
2. Why is audience identification important in this practical?
3. What factors should be considered while selecting visualization?
4. How does a data story differ from raw data presentation?
   How the dataset is important in data Storytelling?
5. Give the name of the dataset used for analysis.
6. Write a code to discover insights in a dataset , includes Dataset, load the Dataset, clean the Dataset, show Popular Locations , Age vs Trip Duration , User Type Analysis and conclude students What pattern do students see?

Practical No.5

Create a video (up to 5 minutes) telling a story on given Incidence. Record a video of yourself speaking, or narrate while showing visual props or sketches, or screencast a PowerPoint presentation, etc. Choose how you present the story. Produce a single video file (formatted as a .mov file)

I. Practical Significance

From a student's perspective, creating a video to narrate a story on a given incident offers invaluable practical benefits. It allows students to develop essential communication skills by articulating ideas clearly and engagingly. Through the process of planning, scripting, and recording, students enhance their creativity and critical thinking as they decide how best to present information visually and verbally. This practical exercise also fosters technical proficiency with tools like video editing software and presentation platforms, preparing students for future academic and professional endeavors where effective multimedia communication is increasingly important.

Through this activity, students learn how to transform data into a compelling narrative that can be easily understood by a wider audience. It enhances their ability to explain insights clearly, maintain audience engagement, and use storytelling techniques such as introduction, problem statement, analysis, and conclusion. Students also gain confidence in presenting their ideas verbally and visually.

II. Course Level Learning Outcome

CO1-Identify the characters in data Storytelling.

CO2-Eliminate clutter and grab audience attention.

CO3-Construct Storytelling for the given incident.

III. Laboratory Learning Outcome

1. Construct a structured data story for a given incident
2. Create and narrate a video presentation using visuals and storytelling techniques
3. Apply appropriate communication style and presentation flow
4. Improve visual communication and audience engagement skills

IV. Relevant Theoretical Background

Video-based data storytelling combines narration, visuals, and structured communication to

present information in an engaging way. Instead of only showing charts or reports, video storytelling helps explain data clearly using voice, visuals, and narrative flow.

A good video story generally includes three parts:

- Introduction → defines the problem or topic
- Body → presents analysis, visuals, and insights
- Conclusion → summarizes findings and recommendations

The effectiveness of a video depends on clarity, pacing, visuals, and audience engagement. Different presentation methods such as narration, slides, screen recording, or visual props can be used depending on the topic and audience.

Visual elements like charts, images, sketches, and animations help the audience understand information easily. A well-structured video story improves communication, creativity, and presentation skills.

V. Equipment and Configuration Required to execute

Computer system with internet access

Mobile phone / Webcam

Microphone (optional)

Screen recording software (e.g., OBS / PowerPoint recorder)

MS PowerPoint / Google Slides

Video editing tool (optional)

VI. Multiple-choice questions (MCQs)

1. What is the main purpose of video-based data storytelling?

A) Data storage

B) Communication of insights

C) Coding

D) Data cleaning

2.Which element is essential in a data story?

A) Random data

B) Structured narrative

C) Complex code

D) File format

3.What should be included in the introduction of a data story?

A) Final result

B) Problem statement

C) Code

D) Dataset

4. Which format is required for submission?

A) .txt

B) .jpg

C) .mov

D) .csv

5. Which method can be used to present the story?

A) Only writing

B) Only coding

C) Video narration or screencast

D) Data storage

VII. Exercises

Create a data story video on **historic event** (e.g. first moon landing)

Procedure to create video:

Project Choice: Select a real-life incident or dataset for storytelling.

1. **Understand the Topic:** Identify the problem, audience, and purpose of the story.
2. **Prepare Script:** Write introduction, body, and conclusion.
3. **Design Visuals:** Create charts, slides, sketches, or supporting visuals.
4. **Choose Presentation Method:** Select narration, screencast, slides, or direct video recording.
5. **Record Video:** Record narration and visuals using recording software.
6. **Edit and Finalize:** Add transitions, titles, and improve clarity.
7. **Export Video:** Save final video in .mov format.

VIII. Practical Related Questions

1. Explain the importance of storytelling in video presentations.
2. What factors improve audience engagement in video storytelling?
3. How do visuals improve communication in videos?
4. Why is script planning important before recording?
5. Compare narration-based and slide-based storytelling approaches.

Practical No.6

Construct a Case study on data storytelling for any Musical/Social App.

Example: Spotify takes the data from our listening habits and spins it into an exciting audio and visual experience

I. Practical Significance

This practical focuses on understanding how data storytelling is applied in real-world digital platforms such as musical and social media applications. Modern apps generate massive amounts of user data, including preferences, behavior, engagement patterns, and usage trends. However, the real value of this data lies in how it is transformed into meaningful insights and presented in an engaging and personalized manner.

Through this practical, students learn how applications use data storytelling to enhance user experience and engagement. For example, platforms like Spotify analyze user listening habits and present them in the form of personalized stories such as yearly summaries, top songs, and listening trends. Similarly, social media platforms use data storytelling to show user activity, growth, and interactions.

Data Story telling for musical or social apps, like Spotify's "Wrapped," enhances user engagement and retention by transforming listening habits into personalized, visually appealing narratives. This approach makes complex data accessible and engaging, encouraging users to interact more with the app and share their experiences on social media. The resulting organic marketing boosts the app's visibility and attracts new users. Additionally, continuous delivery of unique insights fosters a sense of loyalty, keeping users engaged over time.

II. Course Level Learning Outcome

CO2-Eliminate clutter and grab audience attention.

CO3-Construct Storytelling for the given incident.

CO4-Transform Data to Visuals.

III. Laboratory Learning Outcome

1. Analyze data storytelling techniques used in musical/social applications
2. Identify data collection and personalization methods
3. Explore visualization and engagement strategies used by apps
4. Construct a structured case study based on application analysis

IV. Relevant Theoretical Background

Modern musical and social media applications collect large amounts of user data such as listening history, likes, shares, watch time, and user interactions. Data storytelling helps convert this raw data into meaningful and engaging experiences for users.

Applications such as Spotify use personalized storytelling features like yearly summaries, favorite songs, and listening trends to increase user engagement. Similarly, social media apps present user activity and trends through visual dashboards and interactive stories.

Effective data storytelling in apps involves four important steps:

- Data collection
- Data analysis
- Data visualization
- Personalized storytelling

A good case study should explain how data is collected, how insights are generated, what visual techniques are used, and how the storytelling improves user experience and engagement.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Access to apps like Spotify or any social app

VI. Multiple-choice questions (MCQs)

1.What is the purpose of data storytelling in apps?

A) Store data

B) Engage users with insights

C) Delete data

D) Code data

2.Which type of data is used in musical apps?

A) Weather data

B) Listening history

C) Coding data

D) Text files

3.What is a case study?

A) Dataset

B) Detailed analysis of a real scenario

C) Chart

D) Code

4.What makes data storytelling effective in apps?

A) Complexity

B) Personalization

C) Coding

D) Storage

5.Which tool is commonly used for dashboards?

A) Notepad

B) Excel only

C) Power BI

D) Paint

VII. Exercises

Project Choice: Select any musical or social media application.

1. **Understand the Application:** Explore how the app collects and presents user data.
2. **Identify Data Used:** Observe what type of user data is collected.
3. **Analyze Storytelling Features:** Identify how the app converts data into stories or insights.
4. **Study Visualization Techniques:** Observe charts, colors, animations, and layouts used.
5. **Construct Case Study:** Write introduction, analysis, findings, and conclusion.
6. **Present Findings:** Explain how data storytelling improves user engagement.

Additional Real-World Apps (Choose Any One)

1. Instagram (user engagement insights)
2. Facebook (activity analysis)
3. YouTube (watch history insights)
4. Twitter/X (trend analysis)
5. Netflix (watch recommendations)
6. Amazon (shopping behavior)

7. Swiggy/Zomato (order trends)
8. LinkedIn (profile analytics)
9. Snapchat (user interaction data)
10. Google Maps (travel patterns)

VIII. Theory Related Questions

1. Explain how personalization improves data storytelling in apps.
2. How do musical and social apps use visualization techniques?
3. Why is user engagement important in data storytelling?
4. Explain the role of data analysis in personalized recommendations.
5. Compare storytelling approaches used in musical and social apps.
6. Write python code to load dataset for any musical data, display most played song, user listening time, users favourite song etc?

Practical No.7

Implementation of a python program that performs data cleaning on any dataset.

I. Practical Significance

This practical focuses on one of the most important steps in data analysis and data storytelling, which is **data cleaning**. In real-world applications, raw data is often incomplete, inconsistent, or contains errors such as missing values, duplicate records, and incorrect formats. Such data cannot be directly used for analysis or visualization, as it may lead to incorrect insights and poor decision-making.

Data cleaning is practically significant because the quality of data directly impacts the accuracy and reliability of the analysis, machine learning models, and decision-making processes that rely on it. Clean data helps in reducing biases and errors in statistical conclusions and predictions, leading to more robust and actionable insights. For businesses, this can mean better customer insights, improved operational efficiency, and more informed strategic decisions. In research, it ensures that findings are valid and replicable. Overall, clean data is foundational to any data-driven endeavor, ensuring that subsequent analyses are based on reliable information

II. Course Level Learning Outcome

CO2-Eliminate clutter and grab audience attention.

CO3-Construct Story telling for the given incident.

CO4-Transform Data to Visuals.

III. Laboratory Learning Outcome

1. Identify errors and inconsistencies in datasets
2. Apply data cleaning techniques using Python
3. Handle missing values and duplicate records
4. Prepare clean data for visualization and analysis

IV. Relevant Theoretical Background

Data cleaning is the process of identifying and correcting errors in a dataset before analysis or visualization. Real-world data is often incomplete, inconsistent, or contains duplicate values, which can lead to incorrect analysis and poor decision-making.

Common data issues include missing values, repeated records, incorrect formats, and outliers. Data cleaning improves the quality, consistency, and reliability of the dataset.

Python libraries such as Pandas and NumPy are widely used for data cleaning because they provide simple functions to load, inspect, clean, and transform data efficiently.

Typical data cleaning tasks include:

- Handling missing values
- Removing duplicates
- Correcting data types
- Standardizing formats
- Detecting outliers

Clean data is very important because accurate analysis and meaningful data storytelling depend on it.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

Optional tools: Python/ Microsoft Power BI / Tableau

Dataset (CSV file)

VI. Multiple-choice questions (MCQs)

1. What is data cleaning?

A) Data storage

B) Removing errors from data

C) Data visualization

D) Coding

2.Which Python library is commonly used for data cleaning?

A) Matplotlib

B) Pandas

C) TensorFlow

D) HTML

3.What is a missing value?

A) Duplicate data

B) Empty or null data

C) Correct data

D) Numeric data

4.What does removing duplicates do?

A) Adds data

B) Deletes repeated records

C) Changes format

D) Visualizes data

5. Which function is used to check missing values in Pandas?

A) isnull()

B) plot()

C) read()

D) print()

VII. Exercises

Dataset Selection: Select any CSV dataset for cleaning.

1. **Load Dataset:** Import dataset using Pandas.
2. **Inspect Data:** Check missing values, duplicates, and data types.
3. **Handle Missing Values:** Replace or remove null values.
4. **Remove Duplicates:** Identify and remove repeated records.
5. **Correct Data Formats:** Ensure proper formatting of numerical and text fields.
6. **Analyse Cleaned Data:** Verify that the dataset is consistent and usable.
7. **Save Clean Dataset:** Export cleaned dataset for future analysis.

Sample Exercises (Select any one):

1. Hospital records
2. Traffic data
3. Online shopping data
4. Weather dataset
5. Social media data
6. Electricity consumption
7. Transport dataset

VIII. Theory Related Questions

1. Explain the importance of data cleaning in data analysis.
2. What problems can occur due to dirty data?
3. Compare removing and replacing missing values.
4. Why are duplicates harmful in data analysis?
5. Explain how clean data improves visualization and storytelling.
6. Write a python program to Analyze trends and identify issues do dataset profiling, missing value heatmap, duplicate analysis, outlier detection, missing value imputation, correlation analysis?

Practical No.8

Create Bar chart for data visualization using Single distribution.

Example: Histogram of the ages of the train/flight passengers

I. Practical Significance

This practical focuses on visualizing a single distribution of data using bar charts and histograms. In real-world data analysis, understanding how data is distributed is essential for identifying patterns, trends, and anomalies. For example, analyzing the age distribution of passengers in a train or flight can help in decision-making such as service planning, pricing strategies, and customer segmentation.

Visualizing the age distribution of train or flight passengers is crucial for several practical reasons. Firstly, it helps transportation companies understand their customer demographics better, enabling them to tailor services and amenities accordingly. For instance, if a significant portion of passengers are elderly, accessibility and comfort might be prioritized differently compared to a younger demographic where technology and connectivity might be more valued. Secondly, understanding age distribution aids in planning marketing strategies. For example, targeting promotions or discounts based on age groups can be more effective when backed by data on passenger age demographics.

II. Course Level Learning Outcome

CO4-Transform Data to Visuals.

III. Laboratory Learning Outcome

1. Analyze distribution of numerical data using visualization techniques
2. Create histogram and bar chart using Python
3. Interpret trends, spread, and outliers in data
4. Compare distribution characteristics using visualization

IV. Relevant Theoretical Background

A distribution represents how data values are spread across a range. Single distribution analysis focuses on one numerical variable such as age, marks, salary, or temperature. Understanding distribution helps identify patterns, central tendency, variation, and outliers in the dataset.

Histograms are widely used for visualizing continuous numerical data. In a histogram, data is grouped into intervals called bins, and the frequency of values within each interval is represented

using bars.

Key concepts in distribution analysis include:

- Frequency distribution
- Data spread
- Normal and skewed distributions
- Outlier detection
- Bin size selection

Histograms are useful in data storytelling because they help reveal hidden trends and patterns quickly. Choosing an appropriate number of bins is important because too many or too few bins can distort interpretation.

Python libraries such as Pandas, Matplotlib, and Seaborn provide advanced visualization capabilities for analyzing distributions effectively.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system

Python (Jupyter Notebook / Google Colab)

Libraries: Pandas, Matplotlib

Dataset

VI. Multiple-choice questions (MCQs)

1.What is a histogram used for?

A) Storing data

B) Showing distribution of numerical data

C) Coding

D) Writing reports

2.What does a bin represent?

A) Data value

B) Data interval

C) Chart type

D) Code

3.Histogram is used for:

A) Categorical data

B) Continuous data

C) Text data

D) Image data

4.Which library is used for plotting in Python?

A) Pandas only

B) Matplotlib

C) NumPy only

D) HTML

5.What does frequency mean?

A) Data type

B) Number of occurrences

C) Chart style

D) File format

VII. Exercises

Dataset Selection: Select any numerical dataset.

1. **Load Dataset:** Import dataset using Pandas.
2. **Explore Data:** Analyze statistical properties such as mean, median, and standard deviation.
3. **Visualize Distribution:** Create histogram and bar chart.
4. **Analyze Bin Size:** Experiment with different bin values.
5. **Identify Outliers and Trends:** Observe skewness and unusual values.
6. **Interpret Results:** Explain what the visualization reveals about the dataset.

Sample dataset (Select any one):

1. Temperature data
2. Electricity usage
3. Exam scores
4. Population data
5. Height/weight data
6. Sales amount

7. Website visit duration

VIII. Practical Related Questions

1. Explain the importance of distribution analysis in data science?
2. How does bin size affect histogram interpretation?
3. Differentiate between normal and skewed distribution?
4. Why are histograms useful for outlier detection?
5. Compare histogram and bar chart with suitable examples?
6. Write a python program to perform Exploratory distribution analysis in that perform Dataset Profiling, Advanced Histogram, Outlier Detection, Distribution Interpretation, Compare Different Bin Sizes etc?

Practical No.9

Create Bar chart for data visualization using Many distributions.

Example: Histogram of the gender and ages of the train/flight passengers.

I. Practical Significance

This practical focuses on visualizing multiple distributions simultaneously, which is an important step in comparative data analysis. In real-world scenarios, data often contains multiple variables, and analyzing them together provides deeper insights. For example, analyzing age distribution along with gender for train or flight passengers helps in understanding demographic patterns and customer segmentation.

Through this practical, students learn how to compare distributions across categories (e.g., male vs female passengers) using visualizations such as grouped bar charts or multiple histograms. This helps in identifying trends, differences, and relationships between variables.

Understanding the age distribution of passengers is crucial for various operational aspects in transportation. For airlines, it helps in tailoring services such as in-flight entertainment, meal options, and even seat configurations based on the predominant age groups onboard. For railways,it informs decisions on ticket pricing, seating arrangements, and amenities to cater effectively to different age demographics. Moreover, it aids in planning marketing strategies and customer service initiatives tailored to the specific needs and preferences of different age groups.

II. Course Level Learning Outcome

CO4-Transform Data to Visuals.

III. Laboratory Learning Outcome

1. Analyze relationships between multiple variables using visualization
2. Create comparative histograms and grouped charts using Python
3. Interpret patterns across different categories
4. Compare distributions for decision-making and storytellingdistribution.

IV. Relevant Theoretical Background

Multiple distribution analysis involves comparing more than one variable to understand relationships, differences, and trends in the data. For example, analyzing age distribution along with gender helps identify demographic patterns and user behavior.

Histograms and grouped visualizations are commonly used to compare distributions across

categories. These visualizations help reveal patterns such as dominant groups, variation across categories, and hidden trends.

Common techniques for visualizing multiple distributions include:

- Multiple histograms
- Grouped bar charts
- Box plots
- Violin plots

Such analysis is useful in transportation, healthcare, marketing, and business analytics because it supports better decision-making based on comparative insights.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system

Python (Jupyter Notebook / Google Colab)

Libraries: Pandas, Matplotlib

Dataset (Passenger age and gender data)

VI. Multiple-choice questions (MCQs)

1. What is a multiple distribution?

A) Single data value

B) Analysis of more than one variable

C) Dataset

D) Code

2.Which chart is best for comparing groups?

A) Histogram only

B) Grouped bar chart

C) Text

D) Table

3.Gender is which type of variable?

A) Numerical

B) Categorical

C) Continuous

D) Binary only

4.Age is which type of variable?

A) Categorical

B) Numerical

C) Text

D) Label

5.Which library is used for plotting?

A) Pandas only

B) Matplotlib

C) HTML

D) Excel only

VII. Exercises

Procedure

Dataset Selection: Select a dataset containing multiple variables.

1. **Load Dataset:** Import dataset using Pandas.
2. **Explore Variables:** Identify numerical and categorical variables.
3. **Group Data:** Group data based on categories such as gender or department.
4. **Create Comparative Visualizations:** Generate histograms, grouped bar charts, or box plots.
5. **Analyze Patterns:** Compare trends and identify relationships.
6. **Interpret Results:** Explain insights obtained from the visualizations.

Additional Real-World Scenarios (Choose Any One)

1. Student marks by gender
2. Salary by department
3. Sales by region and category
4. Temperature by city
5. Website traffic by device type
6. Hospital data by patient type

VIII. Practical Related Questions

1. Explain the importance of comparative visualization in data analysis?
2. Differentiate between grouped bar chart and multiple histogram?
3. How does multiple distribution analysis support decision-making?
4. Why is grouping important in comparative analysis?
5. Explain challenges involved in visualizing multiple variables?
6. Write a python program to Compare distributions analytically in that first Load Dataset, Compare Gender-wise Distribution, Group-wise Statistical Analysis, Box Plot Comparison, Correlation & Crosstab etc

Practical No.10

Implementation of Python Program for Pie Chart Visualization Using Dataset.

I. Practical Significance

This practical focuses on visualizing proportional data using pie charts, which is an important technique in data storytelling. In many real-world scenarios, it is necessary to understand how different categories contribute to a whole. For example, analyzing market share, customer distribution, product categories, or budget allocation requires representing proportions clearly.

Through this practical, students learn how to load a dataset using Python and represent categorical data as a pie chart. Pie charts help in quickly identifying the percentage contribution of each category, making them useful for presentations and reports.

Implementing a Python program that loads any dataset and plots a pie chart is immensely valuable for data visualization and analysis. Pie charts are a popular tool for representing categorical data, providing a clear visual distinction of how different parts contribute to the whole. This program allows users to quickly and effectively transform raw data into an easily interpretable visual format, aiding in decision-making processes and presentations. Whether it's for business analytics, academic research, or personal projects, being able to generate pie charts from datasets helps in summarizing and communicating complex data insights effectively.

II. Course Level Learning Outcome

CO4-Transform Data to Visuals.

III. Laboratory Learning Outcome

1. Load and analyze categorical dataset using Python
2. Create pie chart visualization for proportional analysis
3. Interpret category-wise contribution using visualization
4. Compare pie chart effectiveness with other visualization techniques

IV. Relevant Theoretical Background

Pie charts are widely used for visualizing proportional or categorical data. They represent how different categories contribute to a complete dataset. Each slice in the chart represents a category, and its size indicates its percentage contribution to the whole.

Pie charts are useful when the number of categories is limited and when the goal is to compare proportions rather than exact values. They are commonly used in business analytics, market

share analysis, customer segmentation, budget allocation, and survey analysis.

However, pie charts should be used carefully because too many categories can make interpretation difficult. In some situations, bar charts provide better comparison and clarity.

V. Equipment and Configuration Required to execute

Equipment Required:

Computer system with internet access

MS Word / Google Docs

Spreadsheet (Excel/Google Sheets)

Optional tools: Python/ Microsoft Power BI / Tableau

VI. Multiple-choice questions (MCQs)

1. What is a pie chart used for?

A) Showing trends

B) Showing proportions

C) Coding

D) Data cleaning

2.What does each slice represent?

A) Data type

B) Category proportion

C) Code

D) Dataset

3.When should pie charts be used?

A) For many categories

B) For proportions of a whole

C) For coding

D) For large datasets

4.Which library is used for plotting pie charts?

A) NumPy

B) Matplotlib

C) HTML

D) Java

What happens if too many categories are used?

A) Better clarity

B) Confusion

C) Faster understanding

D) Better design

VII. Exercises

Dataset Selection: Select any categorical dataset.

1. **Load Dataset:** Import dataset using Pandas.
2. **Identify Categories:** Select categorical variable for proportional analysis.
3. **Calculate Category Distribution:** Compute frequency or percentage of categories.
4. **Create Pie Chart:** Visualize proportional contribution of categories.
5. **Customize Visualization:** Add labels, percentages, and formatting.
6. **Interpret Results:** Analyze contribution and business insights from visualization.

Additional Real-World Scenarios (Choose Any One)

Product category sales

Market share data

Website traffic sources

Budget allocation

Transport usage

App usage categories

VIII. Practical Related Questions

1. Explain the importance of proportional visualization in data analysis.
2. When should pie charts be preferred over bar charts?
3. What limitations occur when too many categories are used in pie charts?
4. How do pie charts support storytelling and business insights?
5. Compare pie charts and bar charts for categorical analysis.
6. Write a python program to Analyze category contribution and business insights in that Load Dataset, Category Distribution Analysis, Advanced Pie Chart, Compare with Bar Chart find Business Insight Generation?